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Ensuring Brand Safety by Using Contextual Text Features: A Study of Text Classification with BERT

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Abstract

When advertisements are placed on web pages, the context in which the advertisements are presented is important. For example, manufacturers of kitchen knives may not want their advertisement to appear in a news article about a knife-wielding murderer. The purpose of the current work is to explore the ability of pre-trained language models on text classification tasks for determining whether the content of a given article is brand-safe, that is, suitable for brand advertising.

A Norwegian-language news dataset containing 3600 news items was manually labelled with negative topics. Five pre-trained BERT language models were tested, including one multilingual BERT and four language models pre-trained specifically on Norwegian. Different training settings and fine-tuning methods were also tested for two best-performing models.

It was found that more structurally complex language models and language models trained on corpora that were large or had larger vocabularies performed better on the text classification task during testing. However, the performance of smaller models is also acceptable if there is a trade-off between the better performance and the time and processing power required.

As far as training and fine-tuning settings are concerned, this work found that for news texts, the initial part of the articles, which often contain the most information, is the optimal choice of parts as input to the model BERT.

Another achievement and contribution of this work was the manual tagging of a Norwegian news dataset on negative topics.

This thesis also points to some possible directions for future work, such as experimenting with different label granularity, experimenting with multilingual controlled training, and training with few samples.

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1 Introduction

Classification is the activity of dividing things into groups for certain needs and according to certain criteria. Text classification is the classification of text into categories according to certain rules. The rules can be set by humans or approximated by algorithms from labeled data.

Today, text classification is widely used as a digital service for identifying customer sentiment (Dawei et al., 2021), analysing the speeches of political leaders and entrepreneurs (Di Ciaccio and Giorgi, 2014), monitoring hate and bullying on social media platforms (Hendrawan et al., 2020) and much more. Traditionally, these tasks have been performed manually, but the proliferation of the internet and the scale of data have led organisations to use text classification models to run their business operations more seamlessly. However, with so many models emerging and so many directions to optimize them in, organizations need to understand which model is better for their tasks, and how to tune the models to better fit their need.

1.1 Motivation

The present work was conducted at the Nordic media company Schibsted. For Schibsted, the largest media group in the Nordics, a potential task is to decide whether the news texts published each day by media under the company's name are appropriate for certain advertisements. For example, a page where an article about a plane crash appears may not be the optimal choice for an airline company to place advertisement on. Handling this task manually may be feasible on a much smaller scale, but is virtually impossible for a large volume of work content of the company. The aim of the present research is to automate the text classification task, using a deep learning approach.

1.2 Context

It could be a difficult problem to define categories for news texts. Fortunately, the Internet Advertising Bureau (IAB) (IAB Tech Lab, 2020) has developed a list of widely-accepted categories for classifying the content of websites and applications. In one of their updates, the IAB tech lab included content with a risk rating. Risk refers to how likely some content is to be harmful, offensive and something an advertiser not generally would want to be associated with. This is intended to give greater control to advertisers who want to take a more nuanced approach, rather than simply dismissing a potential advertising opportunity based on its subject matter alone.

A total of 11 topics are in the *sensitive topics* category, as suggested by the IAB content categories, namely (1) Adult and Explicit Sexual Content; (2) Arms and Ammunition; (3) Crime and Harmful acts to individuals and Society and Human Right Violations; (4) Death, Injury, or Military Conflict; (5) Online Piracy; (6) Hate Speech and Acts of Aggression; (7) Obscenity and Profanity; (8) Illegal Drugs / Tobacco / eCigarettes / Vaping/Alcohol; (9) Spam or Harmful Content; (10) Terrorism; (11) Sensitive Social Issues. These labels were not all applied in the classification task in the present work. The details of the application of the labels will be discussed in Chapter 3.

1.3 Purpose

In the above-mentioned context, the aim of the present work is to test and use deep learning text classification models to classify news texts in Norwegian, labeling each text with the negative topics its content contains, in order to better serve advertisers. The problem is defined as a multi-label text classification task.

There has been related work showing that pre-training models based on large corpora can yield very good results and improvements for text classification (Minaee et al., 2020) and other NLP tasks (Chen et al., 2019; Kaneko and Komachi, 2019; Yang et al., 2019), with the great advantage that we do not need to train a new model from scratch, saving significant resources and time. The present work will focus on one popular pre-trained model, BERT, given its excellent performance in different text classification tasks.

Although BERT has achieved amazing results on many natural language understanding tasks, its potential on classification tasks on Norwegian has not yet been fully explored and the performance of different variations of BERT is yet to be tested on Norwegian news classification tasks. To this end, this study was conducted to answer the following research questions:

- For different pre-trained BERT multilingual models containing Norwegian language and BERT monolingual models on Norwegian, which model performs best in classifying brand-unsafe topics in news texts?
- Which settings and fine-tuning methods could have impact on the models and how should the models be set up?

1.4 Outline

The thesis is structured as follows: Chapter 2 introduces text classification methods (Section 2.1), the development of language modeling (Section 2.2), existing metrics and evaluation methods (Section 2.3) and related works of the current study (Section 2.4). Chapter 3 describes the methodology, including the outline of the experiment pipeline (Section 3.1), details on the process of creating the data set (Section 3.2) and experimental design (Section 3.3-3.5). Chapter 4 presents and discusses the results, and Chapter 5 concludes the work.

2 Background

This chapter provides basic background information about text classification methods, the language models studied in this work, and IAB categories the present work followed.

2.1 Existing Text classification Methods

With the explosive growth of data in the information age, manual classification of text data has become time-consuming and expensive. Due to the huge volume of data, manual classification is inefficient and requires too much time or manpower to achieve effective goals.

Imagine, for example, that the administration of a large social network has to determine whether the text posted by individual users contains offensive content. The number of users is large and it is likely that new texts are being posted at any moment. Identifying these manually would require a lot of manual effort, or the content that users have to post would pile up, leading to longer and longer processing times.

Therefore, the use of machine learning methods for text classification becomes important. Computer processing can effectively avoid above mentioned problems and complete repetitive and tedious tasks, while providing annotated data that is usually consistent and of high quality.

There are two main categories of automatic text classification methods, namely rule-based methods and machine learning methods (Q. Li et al., 2020). In turn, machine learning-based methods can be divided into two types: classic machine learning methods and neural network approaches (Minaee et al., 2020).

2.1.1 Rule-based Text Classification

Rule-based classification approaches (Han et al., 2012) typically use a series of IF-THEN rules to classify texts. The rules are combined to form the structure shown below:

```
if...then  
elseif...then...  
...  
elseif...then...  
else...
```

From top to bottom, the current rule attempts to match the text. If the current rule does not match the text, the next rule is used to match until a rule that can match is found or the rules are exhausted, ending the classification process.

In a rule-based classifier, the rules must satisfy the following two important properties: (1) A rule of mutual exclusion, which means that one text cannot trigger two or more rules, and (2) a rule of exhaustion, which refers to every text in the data set be covered by a rule (Han et al., 2012).

The process of building a classifier is to build a set of rules. There are two types of methods to extract classification rules: (1) Direct method, which extracts classification rules directly from a data set, and (2) indirect method, which extracts classification rules from other classification models, such as decision trees and neural networks (Han et al., 2012).

A common algorithm for the direct method is the sequential covering method (Saian and Ku-Mahamud, 2011). The algorithm extracts the rules of one class at a time, and finishes extracting the rules of one class before proceeding to extract the rules of the next class. Certain principles exist when deciding which class extracts rules first. The high proportion of data contained in the class, or the high rate of misclassification of data in the class, etc. are the principles the rules may be based on.

The classification quality of a rule-based classifier can be measured in terms of coverage and accuracy (Shalabi, 2009). Coverage is the percentage of data that triggered the rule in data set. Accuracy refers to the percentage of data that are correctly classified among those that have triggered the rule.

The characteristics of rule-based classification methods are as follows: first, rule-based classification is often used to produce more descriptive models and its classification performance is comparable to that of decision trees; second, it is suitable for dealing with unbalanced class distributions; third, building a rule-based classifier requires specialized and in-depth knowledge of the data domain being classified; finally, rule-based classification systems are difficult to maintain and do not scale well, because adding new rules may affect the results of pre-existing rules.

2.1.2 Classical Machine Learning-Based Text Classification

The classical machine learning-based text classifier has a relatively simple structure. It consists of two major infrastructures: feature representation and classification model (Q. Li et al., 2020).

The purpose of text feature representation is to transform text into a form that can be more easily processed by computers while reducing the loss of information (Yin et al., 2004). Common text feature representation methods include:

- **Bag-of-Words model (BoW)** (Harris, 1981). A piece of text (e.g., a sentence or a document) is represented by the words it contains, regardless of the grammar and the order of the words. The frequency of word occurrences is used as a feature to train classifiers.
- **N-gram** (Broder et al., 1997). A sliding window of size N is operated on the text according to the words, forming a sequence of "grams" of length N. Language models based on n-grams incorporate information about adjacent words and phrases into a lexicon of representations.
- **TF-IDF** (Salton and McGill, 1983). Term frequency-inverse document frequency is a statistical method for assessing the importance of a word for a document set or one of the documents in a corpus. Word frequency and inverse document frequency are used to model documents.
- **Word2vec** (Mikolov et al., 2013). The word2vec model can map each word to a vector to represent the relationship between words. It uses local contextual information to obtain word vectors.
- **GloVe** (Pennington et al., 2014). Global vectors for word representation is the full title. The vectorized representation of words is performed so that the vectors contain as much semantic and syntactic information as possible between them.

Classical classification models rely on the above feature extraction methods. Classical classification algorithms include the following types.

- **Naïve Bayes** (Manning et al., 2008). This algorithm calculate the probability of B occurring in the case of A according to the following formula:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

In the case of text classification, A represents the features of a piece of text, and B is the category to which the measured text belongs.

- **Support vector machines (SVM)** (Cortes and Vapnik, 2004). It is a machine learning method based on statistical learning theory. The algorithm is based on the principle of structural risk minimization, which compresses the data set into a support vector set and learns to obtain a classification decision function.
- **Hidden Markov model (HMM)** (Wang et al., 2007). HMM studies the transfer relationship between hidden states, and the probability distribution of each state to generate the values of the extrinsic features. In the case of text classification, the hidden state is the category to which the text belongs, while the extrinsic states are the features extracted using the feature representation.
- **Decision Tree (DT)** (Quinlan, 1986). A decision tree is a tree structure, in which each internal node represents a judgment on an attribute, each branch represents the output of a judgment result, and finally each leaf node represents a classification result.
- **Random forests** (Ho, 1995). Random forests is an integrated algorithm consisting of decision trees. When performing classification tasks, text samples enter and let each decision tree in the forest judge and classify it separately. Each decision tree will get a result of its own, and whichever result of the decision trees has the most counts, then the random forest will take that result as the final result.
- **K-nearest neighbors algorithm (KNN)** (Altman, 1992). This algorithm calculates vectors of a new text based on feature representations, and selects the K texts that are most similar to the new text in the training text. Among the K similar texts of the new text, the weight of each class is calculated accordingly. The weights of the classes are then compared and the class with most weight is assigned to the text.

Generally speaking, classic machine learning models are simple models trained to learn predefined feature representations and make predictions accordingly. Some of the limitations are: reliance on hand-crafted features requires tedious feature engineering and analysis to obtain good performance, the strong reliance on domain knowledge when designing features makes the approach difficult to generalize to new tasks, these models cannot take full advantage of the large amount of training data because the features (or feature templates) are predefined.

The classical machine models were used extensively for text classification and other classical NLP tasks (Q. Li et al., 2020) before artificial neural network was applied in this area, but they suffered from the limitations mentioned above. It still serves as basis for the completion of text classification tasks today (Kowsari et al., 2019). In contrast, neural network based text classification models can solve the above problems well and improve the performance of classification models (Minaee et al., 2020). Therefore, the present work focuses on neural network-based models.

2.1.3 Neural Network-Based Text Classification

Neural networks have developed rapidly in recent years, and text classification methods based on neural networks have made great progress (Minaee et al., 2020). The core component of these methods is a machine-learned embedding model, which maps text to low-dimensional continuous feature vectors, thus eliminating the need for handcrafted features.

In this section, some of the best-performing models are presented in chronological order to eventually lead to and introduce Transformer-based model, which is the focus of this work, to compare the problems solved so far with this pre-trained language model and to reflect on its further development.

- Recursive neural network (ReNN) (Goller and Küchler, 1996) based models. Recursive neural networks, which recursively learn the semantic and syntactic tree structure of text without artificially setting artificial features, are a major improvement over shallow networks. Words in the text are treated as leaf nodes of the tree, and all nodes are counted into the father node based on their weights, which are calculated recursively to form a representation of the entire text for predicting category labels. A example study of this model is “Semi-Supervised Recursive Autoencoders for Predicting Sentiment Distributions” (Socher et al., 2011).
- Multi layer perception (MLP) (Van Der Malsburg, 1986) based models. They consist of a three-layer network structure, including: an input layer, a hidden layer containing an activation function, and an output layer, each of which consists of a Full Connection Layer (FCL).

One of the important applications is the introduction of Paragraph Vector (document vector), which was proposed by Dai et al. (2015). In the prediction process of the Continues Bag-of-Words (CBOW) language model, the paragraph vector is introduced to save the paragraph information, and the word vectors of the first three words are averaged or spliced with the paragraph vector and sent to MLP to predict the words in the next position.

- Recurrent neural network (RNN) (Rumelhart et al., 1986) based models. Recurrent neural networks are widely used in the field of natural language processing, and their unique time-series processing is consistent with the way of text reading. The model structure can effectively learn historical and location information, which helps to solve the long-distance dependency problem (Liu et al., 2016).

In the backpropagation process, the parameter update of recurrent neural network depends on the gradient, whose value is calculated by the concatenation of derivatives, and when the derivative value is small, the gradient vanishing problem is prone to cause the neural network parameters cannot be updated effectively.

Long short-term memory (LSTM) (Hochreiter and Schmidhuber, 1997) is an improved model of RNN, which can effectively alleviate the gradient vanishing problem: the information flow is controlled by introducing forget gates, update gates, and output gates, and the information at each time node is represented by cell states. LSTM as a text classifier can effectively capture the association features between context feature words and its forget gate structure can filter invalid information (Zhu et al., 2015).

- Convolution neural network (CNN) (Uang et al., 1994) based models. Convolutional neural networks contain convolutional filters capable of extracting image

features and were first applied to image classification tasks. Unlike RNNs, CNNs are capable of convolving text sequences using different convolutional kernels at the same time. “Convolutional Neural Networks for Sentence Classification” (Kim, 2014) applied CNN for sentence classification and proposed the TextCNN structure, which established the status of CNN networks in the field of text classification, as well as natural language processing.

RNN networks are able to capture the semantic dependencies between words in a text sequence; CNN networks are able to capture the association features between convolutional kernels. Therefore, there are also many studies combining the two network structures (Xue et al., 2017; P. Zhou et al., 2016)

- Attention based models. RNN and CNN lead to a certain uninterpretability of the model because of the presence of hidden states. Bahdanau et al., 2014 proposed the first "attention mechanism" in a machine translation task with good results. Inspired by this, many scholars began to apply attention to text classification tasks. Self-attention mechanism can capture the long-range dependency information in text classification tasks by constructing K, Q, and V to characterize the weight distribution among words in a sentence.
- Graph neural network (GNN) (Scarselli et al., 2009) based models. Deep neural networks (DNNs) such as CNNs are able to achieve good results on regular structured data, but poor results on graph structures with more irregular structures. As attention has been focused on graph neural networks, models based on graph neural networks have performed extremely well in tasks such as semantic role labeling, relational classification, and machine translation by encoding the syntactic structure in sentences. GNN-based models are able to learn the semantic structure of sentences, so GNNs are also applied widely in text classification tasks.

2.2 Pretrained Language Models

After describing the main text classification concepts required for the thesis, this section enters the domain of language modelling (LM), which forms the basis of the bidirectional encoder representation (BERT) model later used in experiments in this work.

A language model is one in which, for any sequence of words, is able to calculate the probability of the upcoming linguistic material (Kuhn and De Mori, 1990). From a text generation perspective, a language model can also be defined in the following way: given a phrase (a word or a sentence), the language model can generate (predict) the next word (Bengio, 2008).

In statistical language models, the probability of a sequence of individual words is decomposed into the product of the conditional probabilities of the individual words in the sequence, and the conditional probability of each word is the probability of that word occurring given its context. See Equation 2.1

$$p(S) = p(w_1, w_2, \dots, w_T) = \prod_{i=1}^T p(w_i | w_1, \dots, w_{i-1}) \quad (2.1)$$

where $p(S)$ is probability for sentence, and w_n stands for the occurrences of words in sentence.

There are two problems with the above language model: the parameter space is too large, and the data is severely sparse. The two problems are solved in n -gram language

models based on Markov's assumption that the next word appears only depending on the word or n words that precede it (Broder et al., 1997). See equation 2.2.

$$p(w_1, \dots, w_T) = \prod_{i=1}^T p(w_i | w_{i-1}, \dots, w_{i-n+1}) \quad (2.2)$$

The n -gram language model treats words as discrete variables and uses one-hot representation (Dunning, 1994), which results in two major disadvantages: there are no semantic relations between word representations, which results in extremely limited generalization ability, and the number of parameters in n -gram model is exponential: in order to reduce the number of parameters, the values of n shall be reduced, which results in bad ability in solving long dependence problems.

Conditional language models is an extension of the language model that calculates the probability of a sequence of words given a condition (D'Sa et al., 2021). If the given condition is another sequence of words, the problem becomes a transformation from a sequence of words to another sequence of words, that is, a sequence-to-sequence problem (Z. Li et al., 2018). The study of sequence-to-sequence models has resulted in a large number of new techniques. A representative sequence-to-sequence model is the Transformer (Vaswani et al., 2017). The Transformer model is characterised by being based entirely on attention mechanisms, using attention to implement the encoding and decoding of sequence-to-sequence models, and the passing of information between encoder and decoder.

Inspired by Transformer, the idea of pre-training a language model based on the encoder or decoder it was proposed (Devlin et al., 2018a). The model is first learned by unsupervised learning using a large scale corpus, i.e. pre-trained; afterwards the model is used for a specific task and the parameters are further tuned by supervised learning using a small amount of annotated data, that is, fine-tuned.

Transformer models are powerful language representations, large scale corpora contain rich linguistic representations (such unlabelled data is readily available), and with training becoming increasingly efficient for large scale deep learning, the resulting pre-trained language models can often effectively represent lexical, syntactic and semantic features of the language. Bidirectional Encoder Representation from Transformers (BERT), a powerful pre-trained language model (Devlin et al., 2018a), have become the current the core technologies of language understanding and language generation.

BERT is a pre-training approach to language representation that divides the building of NLP models into two phases: pre-training and fine-tuning. Fine-training involves fine-tuning the learning on a specific NLP task.

The main innovation of the BERT is that it uses 1) a reduced probability of replacing a small number of words with a mask or another random word when training. The aim is to force the model to increase its memory of the context. 2) A loss is added to predict the next sentence, forcing the model to learn the relationship between sentences (Devlin et al., 2018a).

2.3 Metrics and Evaluation

There are multiple metrics that can be used for evaluating text classification systems. Most of the commonly used evaluation metrics are based on a confusion matrix (Stehman, 1997), which is presented in Table 2.1.

A confusion matrix is a rough evaluation of the prediction results and gives a macroscopic view of the prediction results and the original data. True Positive (TP) means that the sample is indeed positive and the model predicts the sample as positive

		Prediction outcome		total
		p	n	
Actual value	p'	True Positive	False Negative	P'
	n'	False Positive	True Negative	N'
total		P	N	

Table 2.1: Confusion matrix

as well. This is one part of the results that are predicted correctly. True Negative (TN) means that the sample is indeed negative and the model predicts the sample as negative as well. This is the other part of the results that are predicted correctly. False Positive (FP) indicates the case where the sample is actually negative, but the model predicts it to be positive. This is one part of prediction errors, which is also known as a Type I Error. False Negative (FN) indicates the case where the sample is actually positive, but the model predicts it to be negative. This is the other part of prediction errors, which is also known as a Type II Error.

Based on the confusion matrix, some basic and commonly used evaluation matrices can be calculated for text classification models.

- Accuracy / Error rate (Fawcett, 2006). Accuracy indicates the number of samples with correct predictions as a percentage of the total, and error rate represents the percentage of samples with false predictions among the total. The sum of accuracy and error rate is 1.

The formula for calculating accuracy and error rate are defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \text{ Error rate} = \frac{FP + FN}{TP + TN + FP + FN} \quad (2.3)$$

The advantage of accuracy and error rate is that it is simpler and more intuitive (Cohen, 1998). It is usually more applicable when the dataset is balanced. Since accuracy considers both positive and negative samples, it also suits better when each class of labels (positive and negative) is equally important.

- Precision, recall, and F-measure (Powers, 2015). Precision refers to the percentage of samples in which the prediction is positive that do turn out to be positive. See Equation 2.4.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2.4)$$

Higher precision indicates higher quality of predicted positive samples. The model may predicts fewer positive results, but the results contain less negative samples. When the cost of mislabeling a sample that is actually negative as positive is high, but the cost of mislabeling a sample that is actually positive as negative is low, one shall require precision to be high.

Recall is the rate of the samples that are predicted as positive and are indeed positive. See Equation 2.5.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2.5)$$

In the case of this work, the trade-off between precision and recall is actually the comparison of costs between a failed advertisement placement (i.e., the content of the news article in which the advertisement is placed has a negative impact on the advertiser), with a news page with no negative content having no ad placement. The brand image of the client advertiser is seen as a more important factor in this work, so recall has a higher priority than precision in the evaluation of the text classification models.

F-Measure is a weighted summed average of Precision and Recall(Powers, 2015). See Equation 2.6.

$$F = \frac{(\alpha^2 + 1) * \text{precision} * \text{recall}}{\alpha^2 * (\text{precision} + \text{recall})} \quad (2.6)$$

When determining the value of the parameter α , if recall, compared to precision, is the value that should be of more concern, a larger α would be selected.

When the parameter $\alpha = 1$, it represents the same focus on the two indicators of precision and recall. Its score can be said to be the harmonic mean of precision and recall.

- Receiver Operating Characteristic (ROC) curve (Metz, 1978). The ROC curve is based on two metrics, true positive rate (TPR) and false positive rate (FPR). TPR indicates how many of the positive samples are actually predicted to be positive. It is an equivalent to recall. FPR indicates how many of the negative samples are wrongly predicted to be positive. See Equation 2.7.

$$TPR = \frac{TP}{TP + FN}, \quad FPR = \frac{FP}{FP + TN} \quad (2.7)$$

In machine learning models, the output in many cases is the prediction probability. It is necessary to set a classification threshold for the prediction probability, such that a prediction probability greater than the threshold is a positive case, and the opposite is a negative case (Fawcett, 2006). All samples are sorted by their predicted probability, and the corresponding FPR and TPR are calculated using the predicted probability of each sample as the threshold. The data are then represented in a chart with FPR as the horizontal coordinate and TPR as the vertical coordinate. The ROC is obtained by connecting the obtained data points.

The higher the TPR or lower the FPR, the better the performance of the model. Therefore, the closer the ROC curve of the model is to the upper left corner, the better the model performs.

The ROC curve has the following advantages. It is easy to find the effect of any threshold value for the performance of the model. It helps to choose a better threshold value. Performance can be compared for different models. Furthermore, ROC curve is insensitive to data imbalance and can still work well when measuring imbalanced data.

- Area under ROC curve (ROC-AUC) (Hanley and Mcneil, 1983). ROC-AUC is the area under the ROC curve and the horizontal axis. One model is surely performing better than another if the ROC curve of it is completely above the ROC of another. However, sometimes the ROC curves of different models will have intersections and it can be difficult to compare model effects. This is the case when calculating the ROC-AUC is applicable to compare the model effects. Unlike the metrics of accuracy, precision, recall, and F-score, the value of AUC does not depend on the choice of threshold.

ROC-AUC is more practical when the samples are not balanced. This is because ROC-AUC is not sensitive to the proportion of positive and negative samples. It is also the case when positive samples and negative samples are of same concern.

- Precision-Recall (PR) curve. The vertical axis of the PR curve is precision and the horizontal axis is recall. By changing the threshold that distinguishes positive from negative, a series of values of precision and recall can be obtained. The PR curve is a plot of these points on the coordinate system. For the same model, the higher the recall, the lower the precision. The PR curve allows for observing at what recall point precision starts to decline most dramatically. This feature can be used to determine the optimum threshold.
- Area under PR curve (PR-AUC). As with ROC-AUC, the area under the PR curve can also be calculated as a way of describing the performance of the model. The PR-AUC is an equivalent of the average of precision calculated for each recall threshold.

PR-AUC is applicable when selecting a threshold. It also works better when positive samples are of more concern. This is because both Precision and Recall measure the ability of the model to predict positive samples.

2.4 Related Work

The problem of text classification is considered a very classical problem in the field of natural language processing. Research on statistical classification methods can be traced back to the earliest days of the use of Naive Bayes on document classification. In the work of Maron (1961), the computers read documents and determined the category to which they belonged based on the clue words that appeared. Later studies also used KNN for text classification. Cover and Hart (1967) claimed that the information for classification in an infinite sample set is contained in half of the nearest neighbours. Since neighbours have similar behaviour and characteristics, they can be considered as belonging to the same group. Predictions can therefore be made by matching the new text to the closest matches in the training data. Cortes and Vapnik (2004) proposed the idea of using support vector machines (SVMs) for text classification. They used the word stems method to represent documents, i.e. when a word occurs more than three times, it and its occurrences are counted as features. Breiman (2004) made a major contribution to the Random Forest algorithm and applied it to text classification.

All of the above methods are now considered shallow machine learning methods. They are more accurate and stable than the more traditional rule-based text classification and do not require deep knowledge of the domain of the text to be classified in order to construct rules. However, what they all have in common is the need to extract manually created features such as bag-of-words and TF-IDF, and their performance is largely limited by feature extraction.

In 2010s, Deep Learning was introduced into the field of text classification. The Recursive Neural Network (ReNN) (Pouyanfar et al., 2018) are used in text classification

tasks where they can automatically learn the semantics of the text. Alsmadi et al. (2009) presented a method for text classification using Multi-Layer Perceptron, a feed-forward artificial neural network model. The Paragraph Vector (Paragraph-Vec) is one of the most commonly used MLP-based methods. In this method, paragraph vectors are calculated and are also used as input and involved in the prediction process. These applications of Deep Learning to the field of text classification not only reduced the limitations of manual feature extraction, but also led to the best results at the time.

After the first applications of ReNN and MLP, other neural network models were constructed and introduced into the task. Recurrent Neural Network (RNN) (Rumelhart et al., 1986) as neural networks with short-term memory, are often used for sequence construction. When applied to text classification, vanilla's RNNs sometimes do not perform well or even worse than the feed-forward neural networks, while the Long-Short-Term Memory (LSTM) neural network emerge and was combined with RNN (Zhu et al., 2015) that can learn long-term dependency better and thus improve performance. Uang et al. (1994) introduces an unbiased Convolutional Neural Network (CNN) from a CNN network invented for image classification for text classification. Both CNNs and RNNs have achieved considerable success in the field of text classification, but with the introduction of the attention mechanism (Vaswani et al., 2017), more advanced neural networks have been constructed. The attention mechanism is an effective mechanism for identifying relevant words in text. An example of its application is the work of X. Zhou et al. (2016), where a cross-lingual attention-based LSTM approach to sentence meaning classification is presented.

The text classification approaches most relevant to this work are based on the pre-trained language models approach. They learn the semantic representation of global and significantly improve the performance of individual NLP tasks, including text classification tasks.

3 Methodology

This chapter explains the method of obtaining a data set based on the Norwegian language news data provided by Schibsted and how the BERT classification model was trained. As a basis for all classification systems, the BERT model was used (downloaded from huggingface), and the pre-processing, training, testing and validating was done using PyTorch Lightning, as described in more detail below.

3.1 Task

The architecture of the pipeline is outlined as follows: (1) acquisition of the data set, (2) data labeling and pre-processing, (3) GPU training, and (4) experimental setups.

- **Acquisition of the data set:** The Norwegian news data was provided by Schibsted. After a filtering process (described in Section 3.2), a dataset consisting of 3600 news items was formed.
- **Data labeling and pre-processing:** Before training multiple BERT models, each news item in the data set was labeled with the categories described in Section 3.2. They were then converted to the necessary format that BERT requires.
- **GPU training:** All models were trained in the Amazon Web Services, cloud computing, using its GPUs.
- **Experimental setups:** All BERT models are trained on the training dataset and the results are compared. After that the two best performing models were selected and fine-tuned to get the results for comparison.

3.2 Data Sets

The experimental process of this work involved utilizing a dataset consisting of 3600 tagged Norwegian-language news articles obtained from prominent media companies, specifically VG (Verdens Gang), Aftenposten, E24, Stavanger Aftenblad, and Bergens Tidende. These news articles, spanning a five-year period from February 1st, 2017 to January 31st, 2022, were selected based on a defined set of partially negative terms, manually identified for retrieval purposes.

3.2.1 IAB Categories

Before discussing data filtering and labeling, it is necessary to explain the categories of data classification for this work. This work focuses on training text classification models to identify and categorize unseen news articles into specific "unsafe for brands" categories. The team from Schibsted, whose focus is on this area, is using Content Taxonomy 2.2, released by the IAB Technology Laboratory in October 2020, from which also comes the classification followed in this work.

Content Category 2.2 is a classification category that is widely used in the editorial and advertising fields. The aim of it is to quickly reach a consensus that both parties understand what the topic of an article is about. Across all categories and risk levels,

there are 11 topics rated as "sensitive" that require special attention in terms of brand safety. The topics are as follows:

- Adult & Explicit Sexual Content
- Arms & Ammunition
- Crime & Harmful Acts to Individuals and Society and Human Right Violations
- Death Injury, or Military Conflict
- Online Piracy
- Hate Speech & Acts of Aggression
- Obscenity and Profanity
- Illegal Drugs/Tobacco/E-Cigarettes/ Vaping/Alcohol
- Spam or Harmful Content
- Terrorism
- Sensitive Social Issues

The above mentioned are the categories followed in this work. After the categories were identified, some bootstrapping and filtering was carried out in order to select enough suitable news containing content from the above categories to train the model. The categories were translated into Norwegian and a semantic similarity search was conducted for each category with only the headlines of the news items, selecting the top few hundred most similar news articles. Thousands of news articles were then de-duplicated and sorted according to the above topic order, one hundred per topic, repeated as such, and translated by Amazon Web Services (to facilitate the purpose of labeling, and translated texts were not used in model-training etc.), ready to be labeled.

3.2.2 Data Labeling

Each news article can be tagged with more than one of eleven tags, and if no tag is applicable, it would be classified as "none of the above", an exclusion category. If the items themselves are too short, contain only headlines, or are meaningless, they will be categorized as "excluded".

During the course of this study, a comprehensive set of 3,600 meaningful news labels was meticulously compiled. Out of these, 800 labels were jointly marked by the author and the author's supervisor. Notably, there was a substantial overlap between their respective assessments, with only a few instances of disagreement. To ensure accuracy, each divergent label was subject to manual scrutiny, leading to a reevaluation and subsequent determination of the appropriate classification.

It is important to acknowledge the impact of a relatively small dataset on the interpretation of performance scores. With only 3,600 documents available, and approximately 10% allocated for testing, each category comprises a mere 36 documents. Consequently, the significance of individual misclassifications becomes amplified, potentially resulting in notable fluctuations in performance scores. This highlights the challenge of effectively assessing the classification model's performance, particularly in the presence of category imbalance.

There are some findings in the process of labeling that are worth discussing. For some categories, such as *Arms & Ammunition*, *Death Injury, or Military Conflict*, and

Terrorism, semantic similarity search worked well. The bootstrapping and searching process obtained news mostly containing content explicitly related to these topics. These categories themselves have clear meanings, while there is hardly any finer classification under them.

In contrast, for some other labels, they contain more subcategories, which in some cases makes the semantic search process itself more difficult, and the number of news items containing these tags obtained by semantic search is sometimes relatively small, but news items searched by other tags are more likely to be categorized under them as well. These categories include *Illegal Drugs/Tobacco/E-Cigarettes/Vaping/Alcohol* and *Sensitive Social Issues*. The label *sensitive social issues* is particularly problematic, as it is too all-encompassing and can include everything from mental illness, gambling, abortion, to political issues such as immigration. The overly diverse content raises questions about whether the training set of this label would be truly valid.

In contrast to the above two cases, there are some labels that not only rarely appear in the corresponding news articles resulting from a specific semantic search for it, but also rarely appear in the general news. Possible reasons for this situation are, the nature of the news articles causes them not to contain such labels themselves. For example, the content corresponding to *Obscenity and Profanity* will not be, in the first place, a description of the news itself, and secondly, there are fewer news that specifically describe its content, or related to this type of content. A similar situation occurs with the label *Spam or Harmful Content* and *Online Piracy*. Therefore, after discussion, these two tags were removed from the final labeling process and all training and testing processes.

After tagging the news text, all articles with the tags *explicit*, *obscenity and profanity*, *spam or harmful content* and *online piracy* only were eliminated. The remaining articles entered the data pre-processing stage. Noise in the data, such as garbled code, is removed at this stage, and some subheads and content tags included in the news were kept, as they were considered helpful for the training process.

3.3 Experimental Setups

In order to investigate the first research question, which investigates the performance of using different BERT as pre-trained language models in the context of text classification, the following models were selected.

- **BERT multilingual base model (cased)** (Devlin et al., 2018b). This is a model released by Google research. It is a pretrained model on 104 languages with the Wikipedia corpora using a masked language modeling (MLM) objective. This model is case sensitive, has 12 layers, 768 dimensional tensor, 12 attention heads, and 110M parameters in total.
- **NB-BERT-base** (Kummervold et al., 2021), a general BERT-base model built on the large digital collection at the National Library of Norway. It is trained on a wide variety of Norwegian text (both bokmål and nynorsk) from the last 200 years. The structure of this model is the same with BERT multilingual cased model. It has 12 layers, 768 dimensional tensor, 12 attention heads, and 110M parameters.
- **NB-BERT-large** (Kummervold et al., 2021), a general BERT-large model built on the large digital collection at the National Library of Norway. The training data set of this model also includes both bokmål and nynorsk. According to Kummervold et al. (2021), it will give the best results on classification tasks. It has 24 layers, 1024 hidden dimensions, 16 attention heads and 336M parameters.

- **NorBERT** and **NorBERT2** (Kutuzov et al., 2021), a large-scale Transformer-based language model for Norwegian. The models were trained by the Language Technology Group at the University of Oslo. NorBERT corresponds in its configuration to Google’s Bert-Base Cased for English, with 12 layers and hidden size 768.

3.4 Experimental Settings

The BERT models were trained on GPUs with the following settings: batch size 4, 20 epochs, base learning rate $2e-5$, warming up proportion 0.15, and save the best model on the validation set for testing. This setting is based on the work of Sun et al. (2020) in which they experimented and determined the most appropriate parameters for text classification for different BERT models.

3.5 Evaluation Methods

The original data were divided into training, test and validation datasets following the proportions of 80%, 10%, and 10%, and the experimental results were primarily represented on the test dataset.

In the text classification task, it is important to measure the overall accuracy, precision and recall of the models, as well as to obtain the performance of each model without considering the threshold case. Area under ROC curve is suitable for the task. The Area Under the Curve (AUC) is a commonly used performance metric in text classification tasks. It measures the quality of a classification model’s predictions by evaluating the model’s ability to rank instances correctly. The AUC is typically calculated by plotting the true positive rate (sensitivity) against the false positive rate (1-specificity) for various classification thresholds. So, in the evaluation phase, the above-mentioned evaluation approach will be used for each model.

4 Results and Analysis

This section will present the results generated by the experiment.

4.1 Basic Experiments

The maximum length of the BERT input is limited to 512, which according to Devlin et al. (2018a) is for overall computational efficiency. Therefore, in the basic experiments, this work intercepts the first 512 tokens of each news text as training input.

In the training process, the models were trained on the labeled data set to learn whether each news item is positive or negative in each of the nine categories. *None of the above* was also included in the baseline experiment as a separate label and participated equally in the training along with the other labels. Although in real life the number of news articles containing none of the above negative tags would be higher, there is no serious data imbalance under tag *None of the above* because the dataset is bootstrapped with negative tags. After training, the models were made to predict the probability of each news item for each label.

After training the five models, the results of the initial training were derived. To analyze the performance metrics of the classification models at different threshold settings, the AUC-ROC of each model for each label is enumerated in table 4.1 and shown in Figures 1 and 2.

	Nor	Nor2	Multi	NB-base	NB-large	Aver.
A	0.6736	0.9965	0.9965	0.7500	1.0000	0.8833
B	0.7571	0.9929	0.9905	0.8214	0.9833	0.9090
C	0.6541	0.9067	0.8714	0.8282	0.9263	0.8373
D	0.7163	0.9503	0.9360	0.8535	0.9167	0.8746
E	0.5879	0.9939	0.9152	0.6470	0.9864	0.8261
F	0.5296	0.9944	0.9859	0.9211	0.9944	0.8851
G	0.6667	1.0000	0.9453	0.9284	0.9974	0.9076
H	0.7082	0.7790	0.7118	0.7302	0.7888	0.7436
I	0.7858	0.9169	0.9040	0.8837	0.9492	0.8879
Aver.	0.6755	0.9478	0.9174	0.8182	0.9492	

Table 4.1: AUC-ROC results of models and categories.

To simplify the presentation of the graphs and table, each of the nine categories has been replaced by the following letters: A: *Adult & Explicit Sexual Content*. B: *Arms & Ammunition*. C: *Crime & Harmful Acts to Individuals and Society and Human Right Violations*. D: *Death Injury, or Military Conflict*. E: *Hate Speech & Acts of Aggression*.

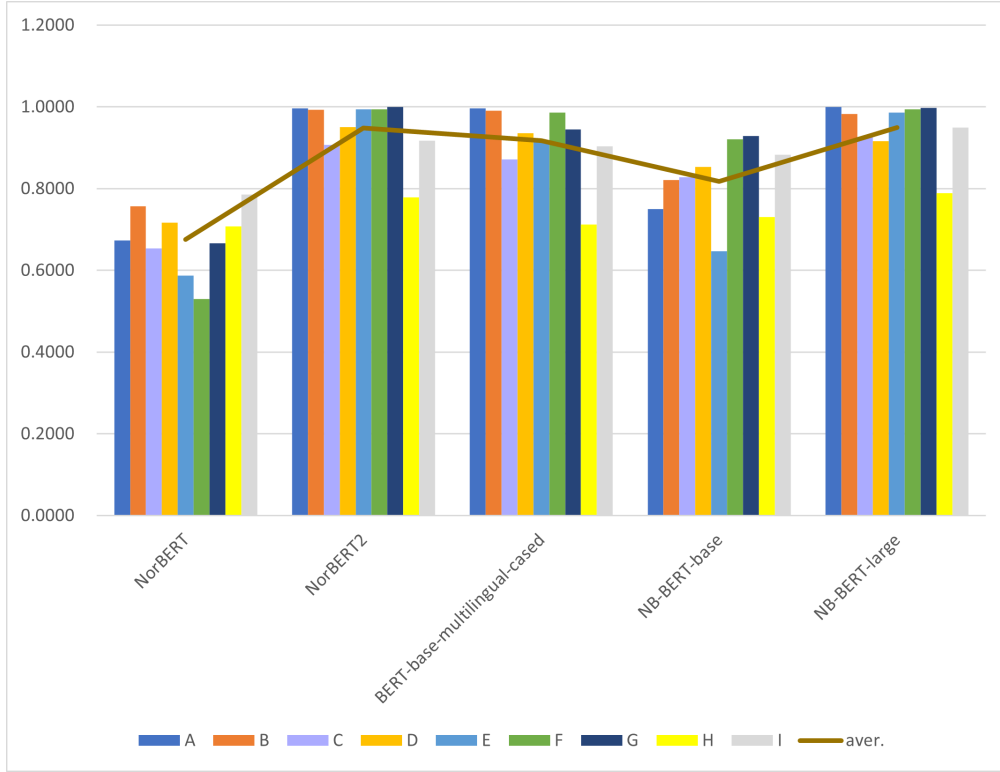


Figure 4.1: AUC-ROC with categories compiled by models.

F: *Illegal Drugs/Tobacco/E-Cigarettes/ Vaping/Alcohol*. G: *Terrorism*. H: *Sensitive Social Issues*. I: *None of the above*.

In table 4.1, the names of the models have been abbreviated in order to shorten the column widths. “Nor” stands for NorBERT, and “Nor2” stands for NorBERT 2. “Multi” is the multilingual cased base BERT model. “NB-base” and “NB-large” are the base and large BERT models trained by National Library of Norway respectively.

4.1.1 Best Performing BERT Models

The table 4.1 and the average curve in figure 4.1 shows that for all categories, considering the average value of AUC-ROC, NorBERT has the worst performance (0.6755) while NB-BERT-large has the best performance (0.9492). This data makes it clear that NB-BERT-large is better able to make a clear distinction for each category than NorBERT. This result is not unexpected and can be explained in two ways: First, the structural complexity of the two pre-trained language models is different. The structure of NorBERT is the same as that of the originally Google-trained BERT-base model, while the structure of NB-BERT-large is the same as that of the originally Google-trained BERT-large model. The latter has twice as many encoding layers as the former, and the number of hidden layers and attention heads is correspondingly greater. The number of parameters is also greatly increased, from 110M to 336M.

According to the work of Jawahar et al. (2019), in a BERT-based language model, the lower layers of the network learn phrase-level information representations, the middle layers learn rich linguistic features, and the higher layers of BERT learn rich semantic information features. It is reasonable to assume that as the number of layers increases, the pre-trained language model is able to acquire a more comprehensive knowledge of the semantics of Norwegian. And in the message text classification task in this work, semantic information is a crucial part of the classification task, so the

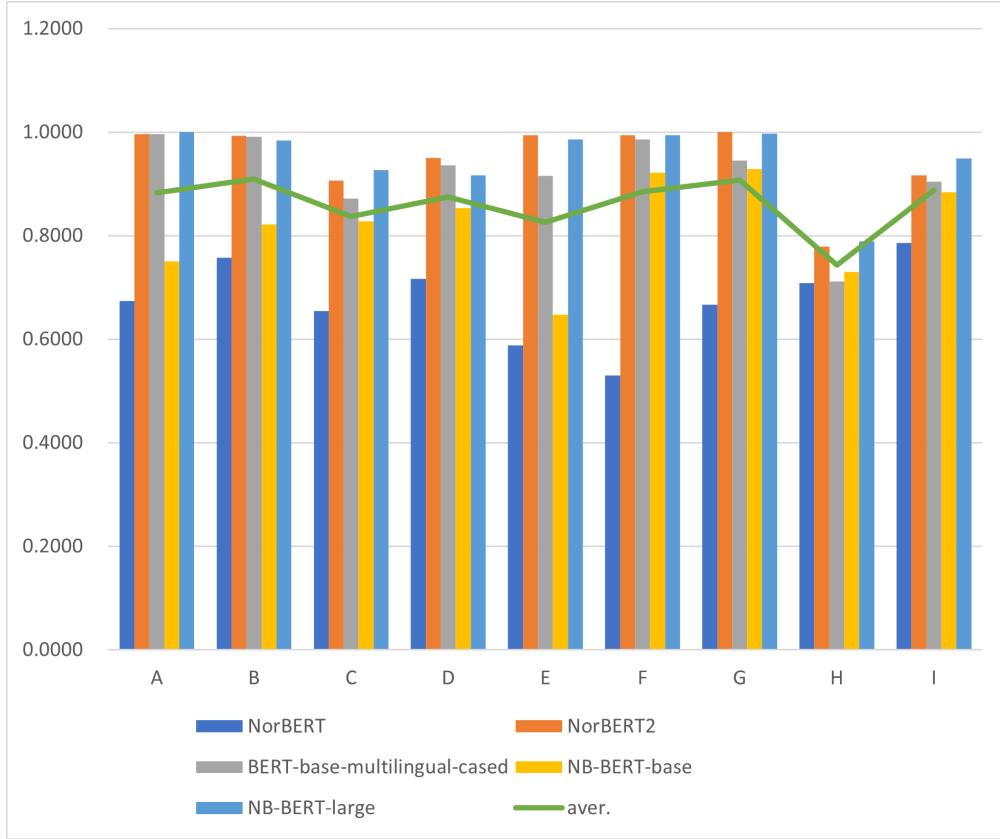


Figure 4.2: AUC-ROC with models compiled by categories.

performance of the more complex model is able to outperform the structural-simpler model.

As for the second best performing model, NorBERT2, it is consistent with the structure of NorBERT. Compared with the latter, it has a larger batch size in both training steps, and has a training corpus size about 7.5 times larger than the latter, and it contains a vocabulary size of 50,000, which is 20,000 larger than NorBERT.

4.1.2 Easiest and Most Difficult Categories to Classify

The data table 4.1 and average curve from figure 4.2 show that category B has the highest AUC-ROC, followed by category G, while the data for category H is the lowest, followed by category E. Category B corresponds to *Arms & Ammunition* and category G stands for *Terrorism*.

Analysis of samples of the news texts within the category of *Arms & ammunition* from the training set shows that certain words are usually present in the texts. The general ones referring to the category include “våpen” (weapon/gun), words that contain the root “våpen”, and “skuddsalver”(gunshot), and words for specific types of guns are usually also present, such as “rifle” and “hagle” (shotgun). This type of feature makes it prominent which category they belong to. It is reasonable to assume that this feature is also present in the test set because of the completely random and equal division of the training and test data sets. The characteristic features of the text in this category could be the reason why it is very easy for the text classification model to distinguish between “positive” and “negative”.

The above conjecture holds true for the category of terrorist attacks as well. Iconic terms such as words that contain the root “terror”: “terrorangrep” (terrorist attack),

“terrordeltagelse” (terrorist participation), “terrorisme” (terrorism); terms referring to some of the political and religious demands that often lead to terrorism, as well as the names of the most common terrorist organisations worldwide, etc.

The next section analyses why the categories H and E, lowest in AUC-ROC, are difficult to distinguish for the language model.

Category H corresponds to *Sensitive Social Issues*. *Sensitive Social Issues* is a very broad category that may include different topics depending on time, country and society. Based on the observations made when labeling the news articles, topics under this category range from relatively personal issues such as abortion and gambling, to general social issues such as employment and public security, to political and government issues and so on. This wide distribution of subtopics may not allow for a sufficient sample size for each subtopic, so the model cannot learn enough features for each subtopic. Furthermore, because there are so many topics that can be classified as *sensitive social issues*, there is also the possibility that there are cases in the test dataset where there are some topics that should also be classified as sensitive social topics, but the model does not have the corresponding knowledge from the training dataset and therefore cannot identify them.

As for category E, *Hate Speech & Acts of Aggression*, it is hard for language models to classify for slightly different reasons. The subjects and objects of hate speech and aggressive behaviour are not clear in the sample under this label in news texts. In a general online environment (i.e. discourse that is not specific to news texts), the texts under this category are usually the hateful, offensive discourse themselves. It would be easier for those classifying this category in texts that are not specific news articles to extract features. But in news texts, the object of hate speech and offensive behaviour under this label is no longer the speakers themselves. It may be a political debate between different political parties, claims in religious sects or inter-ethnic conflicts in the context of historical legacies, etc. The topics of such messages are not only fragmented, their common features are also difficult to identify. When the original data was labeled, many of the news articles categorised in *Hate Speech & Acts of Aggression* are also labeled with *Sensitive Social Issues* at the same time.

4.2 Experiments on Fine-tuning Methods

In training the model, this work also examines the effects of some different settings on the final results.

First, as mentioned in Section 4.1, the tokenized news articles are usually longer than the maximum input length of 512 tokens allowed by BERT models. The initial experiments are based on the first 512 tokens of each news item, including both training and testing data. In this work, the situation when the 512 tokens come from different parts of a news article is investigated, i.e. an experiment with truncation methods is performed.

1. Head-only: keep the first 510 tokens.
2. Tail-only: keep the last 510 tokens.
3. Head+tail: keep the first 255 and last 255 tokens.

NorBERT 2 and NB-BERT-large were re-trained using different truncation methods and the results were observed. The final results of the experiments showed that using head-only configuration would give the best results.

The possible explanation for this result lies in the nature of the news. Some news contain an introduction, usually the first paragraph or the first sentence at the beginning of the news, which briefly reveals the core content of the news. It must write down the most important and exciting facts of the news, outline them, pull the whole text and draw the reader in. Other messages build up "suspense" at the beginning of the news introduction, even if they do not say what happened exactly and directly, the sentences are also enough to mention the categories to which the news belongs.

Second, the predictive ability of different layers of BERT has also been studied. As mentioned before, the representation of linguistic features learned in different layers of BERT is different, so studying whether the knowledge learned in each layer is sufficient for the classification task is a crucial step to improve the efficiency of the model.

The study for the layers of BERT was conducted on NorBERT2, using the full range of categories to ensure accuracy. The results show that all 12 layers of BERT are critical and the AUC-ROC, i.e. discrimination, for each category increases and does not decrease with the number of layers. It reached its highest value in the last shift, so this setting, i.e. all layers of BERT models, was maintained.

5 Conclusions and Future Work

This section summarizes the findings of this work and suggests some possible directions for future research.

5.1 Conclusions

The problem this thesis attempted to solve was to identify in an automated way whether Norwegian-language news was a negative topic and, if so, what it was. This work assumed that using a pre-trained language model trained on a manually labelled dataset can initially serve this purpose. After fine-tuning the model, the language model may be more suitable for this purpose.

Two research questions were posed in this thesis. For the first research question, which of the different pre-trained models performs better in classifying negative topics in news texts, it was concluded that NB-BERT-large performed best. It had the highest differentiation for categories and even reaches 100% for one category. This may be because of its more complex structure, which enabled it to learn more different levels of linguistic knowledge, to acquire semantic knowledge at higher layers that was important for classification problems, and having a richer training corpus, which enabled it to have a greater size of glossary.

But BERT-large also means a higher demand for training time and computing power, as well as a longer prediction time for the results. So if a trade-off is desired, NorBERT₂ is a good choice.

The second research question concerned the improvement or impact of different settings and fine-tuning methods on the results. In this work, it was discovered that for news texts, intercepting the first tokens as input for BERT models was the best option for training and prediction, since the most important content of the news articles was usually elaborated at the beginning. Moreover, the study of the different layers of BERT showed that each layer of BERT learns more language representatives and improves the discrimination for categories.

An additional achievement and contribution of this work is to manually label a data set of Norwegian news on negative topics and use it to experiment with the different kind of BERT models' ability to classify text on negative topics in Norwegian news. A number of settings and fine-tuning measures are experimented with, and corresponding suggestions are given.

5.2 Future Work

In the future work, it would be worth exploring the granularity of the topics. In this work, it has been discovered and discussed that the more detailed the topic (category), the clearer the features in the corresponding text, and the more accurate the model for that subject.

In current work, we are lacking of a meaningful baseline. While the background section of the thesis provides an overview of older methods for solving the main task, it is worth to compare the performance on the current data set. Sklearn module implementing as a baseline will be accomplished in the future work.

It would also be worth investigating whether training the model with more data, performing transfer learning or even multilingual training with Swedish and Danish languages would be helpful in improving the model. It is also possible to train the pre-trained model with further data inside or outside the domain before performing model training.

Finally, to reduce future labelling efforts, it could also be attempted to compare model performance when fine-tuning on fewer examples (few-shot training).

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